A Multifeature TDAO-AIR Clutter Classification Approach for Inshore Ambiguity Identification and Suppression With Azimuth Multichannel SAR System

Chaolei Han, Zhiwei Yang, Qingjun Zhang, Penghui Huang, *Member, IEEE*, and Huanjian Xu

***Abstract*— In the inshore region, the azimuth ambiguity of land clutter is the main factor, which reduces the clutter suppression performance for the multichannel synthetic aperture radar (SAR) system. To address this problem, we propose a multifeature tensor discriminant alternating optimization and affine invariant Riemannian (TDAO-AIR) classification approach. First, the correlation coefficient between channels is obtained to characterize the normalized SAR image amplitude. Next, multilook interferometric processing is performed to reduce the effect of Speckle noise and acquire multilook interferometric complex images. At the same time, we extract the correlation coefficient gradient and the multilook interferometric phase gradient in different spatial directions to characterize the spatial structure. Next, we construct a feature tensor (FT), including correlation coefficient between channels, multilook interferometric phase, and spatial structural gradient information for each range-Doppler unit (RDU). The TDAO algorithm is proposed to extract the core FT (CFT) by optimizing the objective function and getting the projection matrix. Then, the CFT is unfolded along the feature dimension, and the feature covariance matrix (FCM) is constructed for each RDU. With the AIR distance measure, the inshore region is divided into different clutter regions. Finally, in the azimuth ambiguous clutter region, independent and identically distributed (IID) training samples are selected to estimate the clutter covariance matrix (CCM), and robust adaptive clutter suppression is performed in the SAR image domain. The experiments on simulated and real measured data by TerraSAR-X demonstrate that the proposed method can accurately identify and suppress the azimuth ambiguity in the inshore region.**

***Index Terms*— Azimuth ambiguous, clutter classification, clutter suppression, inshore region, synthetic aperture radar (SAR).**

I. Introduction

With the increasing complexity of observation tasks, synthetic aperture radar (SAR) ground/marine moving target indicator (SAR-G/MMTI) technology has attracted much attention, which can detect the ground or marine moving targets and complete velocity estimation while providing high-resolution images [1], [2]. Compared with single-channel SAR systems, multichannel SAR systems exploit the degrees of freedom (DOFs) in azimuth to suppress clutter, which can effectively detect the slow-moving targets submerged in the main-lobe clutter. For unambiguous main-lobe clutter, many methods of clutter suppression and performance analysis have been extensively studied. However, for the inshore region, the clutter energy of the calm sea is usually close to the noise, and the azimuth ambiguous component of strong mainland clutter becomes the main jam for the ship detection. Therefore, how to effectively suppress the azimuth ambiguity of mainland clutter is the key to detect the slow-moving ships in the inshore region.

With the single-channel method, many methods have been proposed to remove the azimuth ambiguity. The first one is preprocessing technology based on optimal antenna design, which reduces the azimuth ambiguous clutter power by the optimal antenna sidelobe [3], [4]. Another one is postprocessing technology-based receiving data, including in-phase cancellation and selective filtering methods. The in-phase cancellation techniques attempt to acquire the great in-phase canceling result of the ambiguous clutter by reconstructing the signal sources accounted for the aliased signal [5], [6]. It can effectively remove the azimuth ambiguity of the strong point targets. However, for the distributed target, it has a bad performance because of the unrecoverability of the high-frequency components in the reflectivity spectra. Chen *et al.* [7] propose a processing framework to suppress azimuth ambiguity by analyzing the ambiguity factor, which

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can commendably achieve in-phase cancellation. However, it severely relies on the precise system parameters and the known location of the azimuth ambiguity. For the selective filtering technique, although it can remove the azimuth ambiguity effectively by exploiting bandpass filters, it unavoidably will reduce the signal-to-noise ratio (SNR) and the azimuth resolution [8]. Besides, the additional speckle noise will be introduced. Guarnieri [9] proposes an adaptive Wiener filter to deal with the aforementioned problems. Martino *et al*. [10] introduce an "asymmetric mapping and selective filtering" (AM&SF) method to further solve the problem that it may not identify ambiguity for some antenna pattern and PRF values. In addition, the image selective restoration (IMSR) [11] and the time-frequency analysis techniques [12] may be used to identify and remove the azimuth ambiguity.

For the multichannel SAR system, the preprocessing and postprocessing technologies are also the typical methods to deal with the azimuth ambiguity. The preprocessing technology is mainly based on digital beamforming (DBF) technology, which utilizes multiple independent received antennas aligned in azimuth to increase the spatial sampling and can recover the unambiguous Doppler spectrum [13], [14]. The postprocessing technology is based on the detection of moving targets with the multichannel SAR image. The most critical step is the location determination of the azimuth ambiguity. Shu *et al*. [15] jointly utilize the interferometric amplitude and phase characteristics of SAR images to choose the clutter classification threshold to determine the position of the azimuth ambiguity. This method requires that the interferometric phase and amplitude histograms must have clearly distinguishable multipeak characteristics, and the threshold parameters need to be manually chosen, which severely restricts the applicability in actual work. Xu *et al.* [16], [17] propose a multifeature autosegmentation-based (MFAutoB) algorithm using SAR image amplitude and interferometric phase between channels, which establishes a limited elementary hybrid model about the multilook interferometric covariance matrix and achieves automatically clutter segment by the expectation maximum (EM) algorithm. Although the MFAutoB method avoids manual selection of the classification threshold, the utilized EM algorithm cannot effectively constrain the number of classified clutter categories. The number of clutter categories is too small to accurately classify the various clutter regions. On the contrary, similar clutter will be divided into multiple categories, and subsequent manual selection is still required. More importantly, the MFAutoB method only exploits the amplitude and phase information, which has a certain limitation to the classification accuracy.

In fact, in the inshore region, the clutter texture in different clutter regions is also different in addition to the amplitude and phase of the SAR image. Generally, the mainland clutter is mainly composed of buildings, vegetation, mountains, and other features. The range-Doppler unit (RDU) has strong spatial correlation information that macroscopic performance is the unique texture characteristics of various features. The azimuth ambiguous clutter is mainly composed of the sea clutter and the azimuth ambiguous component of the mainland clutter. Although the texture of the mainland clutter is still retained, due to the azimuth antenna pattern in the sidelobe and the influence of the sea clutter, the clutter texture is not exactly the same as the land clutter. As for the sea clutter, the clutter texture is random in all directions, which is quite different from the mainland clutter and azimuth ambiguous clutter.

In recent years, feature extraction has been widely used in polarimetric SAR (PolSAR) processing and hyperspectral image classification. In the field of PolSAR processing, feature extraction is mainly dedicated to the acquisition of multifeature vectors, which can be seen as a multifeature matrix operation. Tran *et al.* [18] directly classify with the polarimetric covariance matrix by extracting the polarimetric features of the SAR image, which cannot obtain a satisfactory classification result. Zhang *et al*. [19] utilize the gray level co-occurrence matrix (GLCM) to describe the spatial texture information of PolSAR images, which improves the classification accuracy. Uhlmann and Kiranyaz [20] apply the color information features to the classification of PolSAR images. The classification accuracy gets better to a certain extent. In order to further improve the classification accuracy, many scholars have effectively integrated multiple features of PolSAR images. Akbarizadeh and Rahmani [21] propose a PolSAR clutter classification method that combines texture information and color information and makes full use of the three features, including polarization, texture, and color, which greatly improves the classification accuracy. Pham [22] fuses each polarimetric component of the PolSAR image with structural gradient features to construct an FCM. Then, the AIR distance between the FCMs is exploited to segment the PolSAR image. For the hyperspectral image (HSI) classification, feature extraction is mainly dedicated to reduce data dimensionality. Many methods transform high-dimensional data tensors to low-dimensional spaces based on training samples, which is a tensor rank reduction operation [23]. The most representative methods based on tensor feature extraction include the principal component analysis (PCA) method [24] and the linear discriminant analysis (LDA) method [25]. The main purpose of PCA is to maximize the variance between the converted data, while LDA aims to maximize the distance between different categories. Recently, some improved methods are widely used in hyperspectral space-frequency feature extraction, such as the multilinear principal component analysis (MPCA) method proposed in [26] and the tensor discriminative locality alignment (TDLA) method proposed in [27]. Here, it should be pointed out that the feature extraction method based on tensor can be regarded as an extension of the feature vector method, which further extracts the core feature tensor (FT) (CFT) and can more accurately characterize the feature information of each RDU.

The above tensor-based methods can extract valuable features and reduce the data dimensionality while preserving the local spatial structure, but they ignore the correlated properties of different features in the subsequent classification step by the Frobenius-norm distance between CFTs. They are named the Tensor-FR methods in this article. However, the covariance-based descriptor is able to take those correlated properties into consideration by containing both the variance

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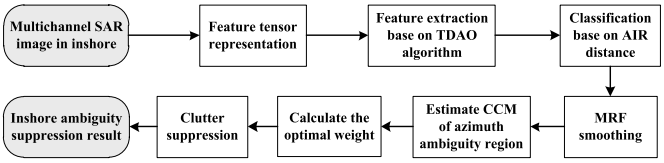


Fig. 1. Framework of the method proposed.

of each feature (i.e., matrix's diagonal entries) and their corrections (i.e., matrix's off-diagonal entries), which can help to boost the distinctiveness of the extracted features. In addition, the tensors of the preceding tensor-based methods are generally constructed by amplitude features with different images independently, which ignores the correlated properties of different channels or spectra maps. In fact, the correlation coefficient and interferometric phase are very valuable features for the multichannel SAR system, which can represent different clutter to a certain extent in the inshore region.

Inspired by the PolSAR image and HSI classification methods, we propose a multifeature TDAO-AIR classification approach for inshore ambiguity identification and suppression with a multichannel SAR system. The framework of the method is shown in Fig. 1. For exploiting correlated properties of SAR images of different channels, the correlation coefficient between channels is first calculated to characterize the normalized SAR image amplitude. Meanwhile, multilook processing is performed on SAR images to reduce the influence of coherent Speckle noise and obtain multilook interferometric complex images. Next, we fuse the correlation coefficient, multilook interferometric phase, and structural gradient information into FT for each RDU. Then, the TDAO algorithm is used to construct an objective function containing core feature information and perform alternate optimization to obtain the CFT. Furthermore, the CFT is expanded along the feature dimension, and the FCM is obtained of each RDU. With the AIR distance metric, the inshore clutter is classified, and the location of the azimuth ambiguity is accurately obtained. Then, the IID clutter samples are selected to estimate CCM. Finally, the optimal weight is calculated, and the adaptive clutter suppression is performed in the azimuth ambiguous clutter region, thereby obtaining a great clutter cancellation result. The main contributions of this article are threefold.

1) The tensor representation is used to extract the SAR image features of the azimuth multichannel system. The correlation coefficient, multilook interferometric phase, and structural gradient information are fused into the FT of each RDU for the first time in the inshore ambiguity identification and suppression field.
2) The proposed TDAO-AIR classification approach further improves the classification performance of the Tensor-based methods by the AIR classifier, which can not only preserve the local structure and the multifeature information to reduce data dimensionality but also take the correlated properties of different features into consideration. This method can enhance the distinctiveness of the extracted features and obtain great classification results.

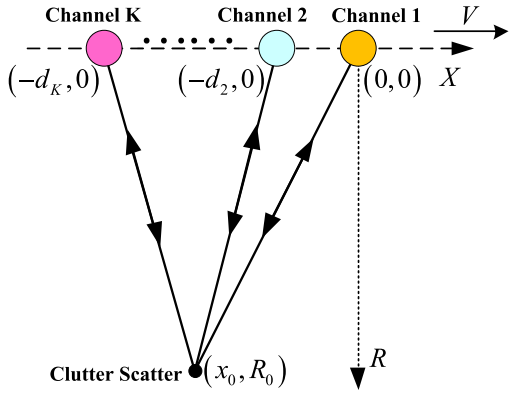


Fig. 2. Geometry of the multichannel SAR system.

The ambiguity suppression method is provided for the azimuth multichannel system. With the help of the inshore ambiguity identification results, the IID ambiguity clutter samples are selected to estimate CCM of ambiguity region, and the great ambiguity suppression performance can be obtained comparing with the traditional ambiguity identification and suppression method.

The rest of this article is organized as follows. Section II gives an introduction to the signal model with multichannel SAR systems. In Section III, the proposed method is described in detail. Some simulated and real measured experiments are presented in Section IV. Finally, Section V concludes this article.

II. Signal Model

For multichannel SAR systems, we assume that there are K receiving channels that are aligned azimuth, as shown in Fig. 2. The radar platform flies along the azimuth direction that is denoted as the x-axis with a constant velocity V. The slant-range direction is orthogonal to the x-axis and denoted as the R-axis. The first channel is considered the reference channel. The effective phase center separation (EPCS) of the kth channel is d_k, and $d_1 = 0$. At time $t = 0$, we assume that the reference channel is located at (0,0), and the clutter scatter point is at coordinate (x_0, R_0).

At time $t = t_m$, the instantaneous slant range between the clutter scatter point (x_0, R_0) and the kth azimuth receiving channel can be represented as

$$R_c^{(k)}(t_m; x_0, R_0) = \sqrt{R_0^2 + (V \cdot t_m - x_0 - d_k)^2} \approx R_0 + \frac{V^2}{2R_0} \cdot \left(t_m - \frac{x_0}{V} - \frac{d_k}{V}\right)^2. \quad (1)$$

Because the radar echo is acquired by the sample at pulse repetition frequency (PRF) in azimuth, the azimuth focused SAR image of the kth azimuth receiving channel is given by [16]

$$s_{c_\text{out}}^{(k)}(t_m; x_0, R_0) = \sum_{\zeta} \sigma_c^{(\zeta)} \cdot \text{sinc}\left[\text{PRF} \cdot \left(t_m - t_0 - \frac{\zeta \cdot \text{PRF}}{k_a} - \Delta t_k\right)\right] \times \exp\left(j \cdot \pi \cdot \frac{(\zeta \cdot \text{PRF})^2}{k_a} + j \cdot 2\pi \cdot \zeta \cdot \text{PRF} \cdot \Delta t_k\right) \quad (2)$$

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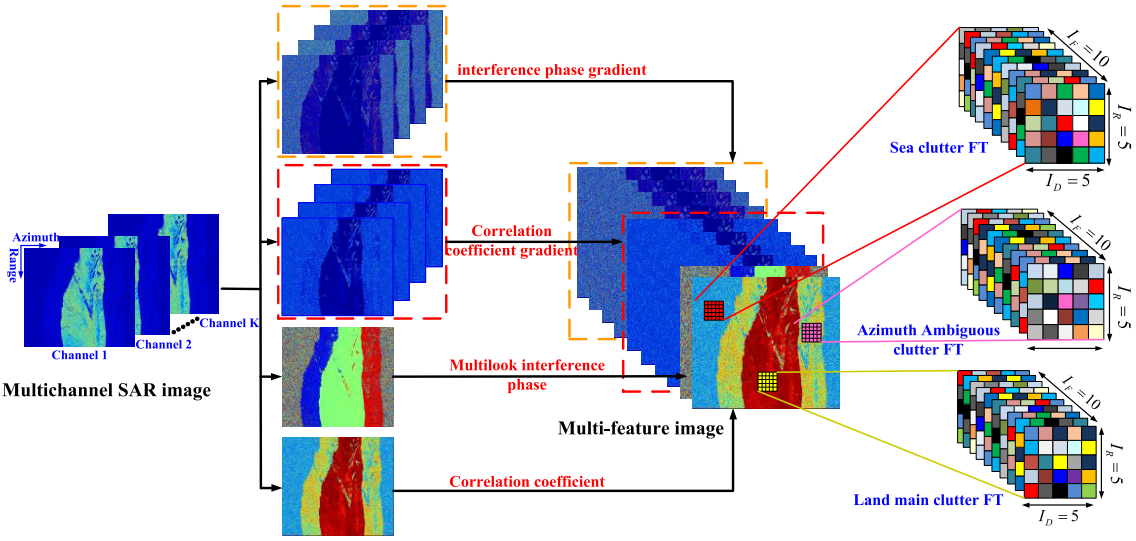


Fig. 3. Multifeature tensor representation of the multichannel SAR image data in the inshore region.

where $k_a = 2V^2/\lambda R_0$ is the azimuth Doppler frequency modulation rate, $t_0 = x_0/V$ is the delay time of scatter point in zero Doppler time, and $\Delta t_k = d_k/V$ is the delay time that the kth receiving channel relative to the reference receiving channel. ζ is the number of the folded azimuth spectrum, and $\sigma_c^{(\zeta)}$ is the clutter scatter amplitude corresponding to the ζth ambiguity clutter. It is obvious that, compared with no azimuth ambiguity, the focused image of the ζth ambiguity region will appear at the position of $t_m = t_0 + \zeta \cdot \mathrm{PRF}/k_a + \Delta t_k$, and the interferometric phase between the kth channels relative to the reference receiving channel has an excess phase offset $2\pi \cdot \zeta \cdot \mathrm{PRF} \cdot \Delta t_k$. In the inshore region, the magnitude and the interferometric phase are diversified among different clutter regions. On the one hand, the magnitudes of the mainland clutter and the ambiguous land clutter component are often larger than the calm sea clutter since the clutter backscattering coefficient of the main-lobe land is much larger than that of the calm sea region. On the other hand, the mean value of interferometric phase between channels is close to zero for the mainland clutter. For the ζth ambiguity, the mean value of interferometric phase between channels is close to $2\pi \cdot \zeta \cdot \mathrm{PRF} \cdot \Delta t_k$, while the interferometric phase between channels obeys uniform distribution with $(-\pi, \pi]$ for the sea clutter. It is obvious that the clutter suppression performance will be degraded because the distributions of magnitude and interferometric phase between channels are various for different regions.

III. Proposed Inshore Ambiguity Clutter Suppression Method

In this section, the proposed method is described in detail. In this article, the method can be divided into two main aspects: clutter classification is based on feature extraction and multichannel adaptive clutter suppression. For clutter classification, we first give a feature description, which mainly carries out multiple features and represents the inshore clutter by tensor, as shown in Fig. 3; Then, the TDAO-AIR algorithm is proposed to extract CFT and perform classification with the AIR distance, as shown in Fig. 4. Finally, under the guidance of clutter classification results, IID-compliant clutter samples are selected for CCM estimation, and adaptive clutter suppression for azimuth ambiguity is completed.

A. Feature Describe

1) Correlation Coefficient and Multilook Interferometric Phase: The SAR image amplitude is the scattering intensity of electromagnetic waves from various clutter points in the radar observation scene, which is the most direct feature to characterize different types of features. In the inshore region, the amplitude of the azimuth ambiguity is the azimuth ambiguous image of the mainland clutter, which is different from the mainland clutter and sea clutter and is an important feature for clutter discrimination. However, in the actual work, the SAR image amplitude is often distributed in a large range and is susceptible to noise. Directly using the gray of the SAR image often causes the problem that clutter classification performance is degraded [28]. For the multichannel system, the correlation coefficient between channels can be regarded as the normalized amplitude of the SAR image, and the amplitude fluctuation range is controlled within the range of 0–1. Moreover, the local smoothing window is used when calculating the correlation coefficient, which can effectively suppress the speckle noise. Therefore, in this article, the correlation coefficient is exploited to characterize the amplitude feature of different clutters in the inshore region.

For an RDU with coordinate (x, y), the correlation coefficient between any two adjacent azimuth receiving channels (channel r and channel q, and $q = r + 1$) can be expressed as follows:

$$\rho_{r,q}(x, y) = \frac{E\{\boldsymbol{e}_r(x, y)\boldsymbol{e}_q^*(x, y)\}}{\sqrt{E\{\boldsymbol{e}_r(x, y)\boldsymbol{e}_r^*(x, y)\}E\{\boldsymbol{e}_q(x, y)\boldsymbol{e}_q^*(x, y)\}}} \quad (3)$$

where $\boldsymbol{e}_r(x, y)$ represents the data vector formed by the RDUs in the data window centered in (x, y) on the rth channel SAR

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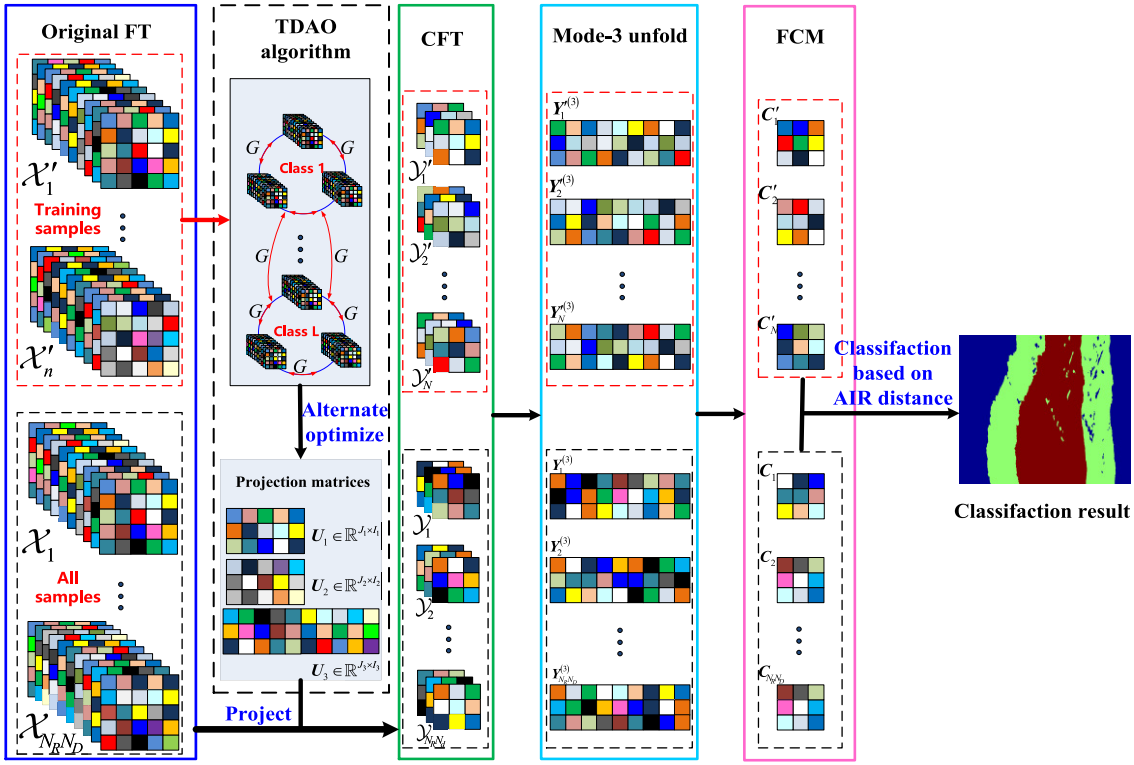


Fig. 4. Framework of clutter classification based on the TDAO-AIR algorithm.

image. $E\{\cdot\}$ is the averaging operation, and $[\cdot]^*$ is the complex conjugate operation. In the inshore region, the mainland clutter has the highest correlation, which is close to 1. The sea clutter has no correlation between channels because it is similar to noise, and the correlation coefficient is close to 0. The correlation for azimuth ambiguous clutter is between the mainland clutter and the sea clutter.

After image registration, the spatial multilook processing is performed to obtain multilook interferometric complex data, including amplitude and phase information of multichannel SAR images. The multilook interferometric data of the channels r and q for the pth RDU can be expressed as follows:

$$\gamma_{r,q}(x,y) = \frac{1}{W^2}\sum_{p\in\Omega_{x,y}} z_r(p)z_q^*(p) = \eta(x,y)\cdot\exp[j\varphi(x,y)] \tag{4}$$

where $\Omega_{x,y}$ is the multilook window domain of the RDU located in (x,y), W^2 is the total number of RDUs in the multilook window, $\eta(x,y) = |\gamma_{r,q}(x,y)|$ is the multilook interferometric magnitude, $\varphi(x,y) = \text{angle}(\gamma_{r,q}(x,y))$ is the multilook interferometric phase, and $z_r(p)$ is the complex value of the pth RDU, which is located in (x,y) on the single-look SAR image of the rth azimuth receiving channel.

The above discussion is based on only two azimuth receiving channels in an SAR system. For a multichannel system (the channels are evenly arranged along azimuth), there is a small fluctuation of the correlation coefficients and interferometric phases for two arbitrary adjacent channels because of the channel inconsistency. In order to minimize this bad effect, the normal correlation coefficient and normal interferometric phase of arbitrary two adjacent channels are used as multiple samples, and the final values are characterized by the mean of the whole samples as follows:

$$\tilde{\rho}(x,y) = \frac{1}{K-1}\sum_{k=1}^{K-1}\rho_{k,k+1}(x,y)$$
$$\tilde{\varphi}(x,y) = \frac{1}{K-1}\sum_{k=1}^{K-1}\varphi_{k,k+1}(x,y) \tag{5}$$

where K represents the number of receiving channels.

2) Spatial Structure Gradient: Gradient information is a very important parameter for the representation and analysis of the structure characteristics of the image. It has been widely used in the fields of computer vision [29], PolSAR image classification [30], and hyperspectral image processing [31]. For the multichannel system, image spatial structure gradient mainly includes amplitude gradient and interferometric phase gradient, where the amplitude gradient can be represented by correlation coefficient gradient, which is more robust for classification. Image gradient not only contains the amplitude and phase characteristics of each RDU but also takes the local correlation between RDUs into account. Different types of features have different local gradients in the spatial location. In order to synthesize the gradient information in different directions of the image, the spatial structure gradients along four different directions are defined to describe the texture information of the multichannel SAR image.

For the RDU with the spatial coordinate (x,y), the corresponding correlation coefficient between channels is $\tilde{\rho}(x,y)$. $\tilde{\rho}'_1$, $\tilde{\rho}'_2$, $\tilde{\rho}'_3$, and $\tilde{\rho}'_4$ are the gradients along with four different

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spatial directions that constitute the correlation coefficient gradient vector as follows:

$$\tilde{\boldsymbol{\rho}}'(x,y) = [\,\tilde{\rho}'_1(x,y) \quad \tilde{\rho}'_2(x,y) \quad \tilde{\rho}'_3(x,y) \quad \tilde{\rho}'_4(x,y)\,]^T \tag{6}$$

where

$$\begin{aligned}
\tilde{\rho}'_1(x,y) &= |\tilde{\rho}(x+r_x,y) - \tilde{\rho}(x-r_x,y)| \\
\tilde{\rho}'_2(x,y) &= \sqrt{2}\cdot|\tilde{\rho}(x+r_x,y+r_y) - \tilde{\rho}(x-r_x,y-r_y)| \\
\tilde{\rho}'_3(x,y) &= |\tilde{\rho}(x,y+r_y) - \tilde{\rho}(x,y-r_y)| \\
\tilde{\rho}'_4(x,y) &= \sqrt{2}\cdot|\tilde{\rho}(x-r_x,y+r_y) - \tilde{\rho}(x+r_x,y-r_y)|
\end{aligned} \tag{7}$$

where r_x and r_y are the radii of the window in the image spatial domain, and $|\cdot|$ is the absolute value calculation operation.

For the multilook interferometric phase $\tilde{\varphi}(x,y)$, the phase gradient vector of RDU whose spatial coordinate is (x,y) can be expressed as

$$\tilde{\boldsymbol{\varphi}}'(x,y) = [\,\tilde{\varphi}'_1(x,y) \quad \tilde{\varphi}'_2(x,y) \quad \tilde{\varphi}'_3(x,y) \quad \tilde{\varphi}'_4(x,y)\,]^T \tag{8}$$

where

$$\begin{aligned}
\tilde{\varphi}'_1(x,y) &= |\tilde{\varphi}(x+r_x,y) - \tilde{\varphi}(x-r_x,y)| \\
\tilde{\varphi}'_2(x,y) &= \sqrt{2}\cdot|\tilde{\varphi}(x+r_x,y+r_y) - \tilde{\varphi}(x-r_x,y-r_y)| \\
\tilde{\varphi}'_3(x,y) &= |\tilde{\varphi}(x,y+r_y) - \tilde{\varphi}(x,y-r_y)| \\
\tilde{\varphi}'_4(x,y) &= \sqrt{2}\cdot|\tilde{\varphi}(x-r_x,y+r_y) - \tilde{\varphi}(x+r_x,y-r_y)|.
\end{aligned} \tag{9}$$

In fact, in an SAR image with 2-D space, each RDU has four pairs of opposite directions, that is, eight gradient directions. However, for azimuth ambiguity identification in the inshore region, it is enough that the above four different gradient directions represent the spatial structure of RDU. For reducing computational complexity, in this article, we only use the absolute value of the gradient in each pair of gradient directions as the gradient feature in all directions.

B. Feature Tensor Representation

For multidimensional data that contained multiple features, the tensor is a good representation tool [32], [33]. It treats multidimensional data as an entirety and can represent each RDU with a basic assumption that RDUs in each patch tensor belong to the same class. In this article, the correlation coefficient (normalized amplitude) between channels, interferometric phase, and spatial structural gradient information of multichannel SAR images are fused together to construct a multifeature image $\boldsymbol{F}$ as follows:

$$\boldsymbol{F} = \{\boldsymbol{f}_i \in \mathbb{R}^{10} | \boldsymbol{f}_i = [\tilde{\rho} \quad \tilde{\varphi} \quad (\tilde{\boldsymbol{\rho}}')^T \quad (\tilde{\boldsymbol{\varphi}}')^T]^T\} \tag{10}$$

where $\boldsymbol{f}_i$ is the multifeature vector extracted of ith RDU, and the size is I_F. Assuming that the SAR image contains $N_R N_D$ RDUs, the FT $\mathcal{X}_i \in \mathbb{R}^{I_R \times I_D \times I_F}$ of the ith RDU is obtained by data selecting with a sliding space window of size $I_R \times I_D$. Then, the FTs' set of all RDUs for the multifeature image $\boldsymbol{F}$ is given as follows:

$$\mathcal{X} = \{\mathcal{X}_i\}, \quad (i = 1, 2, \ldots, N_R N_D). \tag{11}$$

The Tucker decomposition is a principal component analysis method for high-order tensors and can be applied to the field of low-rank tensor estimation [34]. The tensor algebra definitions that are relevant to this article are shown in the Appendix. For any tensor $\mathcal{X}_i \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$, we donate that M is the mode, and the dimensionality of mth mode is I_m. For the core tensor $\mathcal{Y}_i \in \mathbb{R}^{J_1 \times J_2 \times \cdots \times J_M}$, the dimensionality of the mth mode is J_m. The purpose of feature extraction is to find M (in this article $M = 3$) projection matrices $\{\boldsymbol{U}_m \in \mathbb{R}^{J_m \times I_m}, J_m \le I_m, m = 1, 2, \ldots, M\}$ to project any tensor $\mathcal{X}_i \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ to the core tensor $\mathcal{Y}_i \in \mathbb{R}^{J_1 \times J_2 \times \cdots \times J_M}$ as follows:

$$\mathcal{Y}_i = \mathcal{X}_i \times_1 \boldsymbol{U}_1 \times_2 \boldsymbol{U}_2 \times_3 \cdots \times_M \boldsymbol{U}_M, \ i = 1, 2, \ldots, N_R N_D \tag{12}$$

where the operator $\mathcal{X} \times_m \boldsymbol{U}_m$ is the m-mode product and every projection matrix is an orthogonal matrix. It is clear that the data dimensionality of the mth mode of the original FT $\mathcal{X}$ can be reduced from I_m to J_m. Generally, the clutter classification algorithms are based on the CFTs for improving computing efficiency. The extracted core tensors must have two properties: maintain the similarities of the RDUs that belong to the same class and effectively separate the RDUs that belong to the different classes. Thus, it is crucial to accurately and effectively extract the CFT of each RDU for the clutter classification.

C. Feature Extraction Based on TDAO Algorithm

Tensor has been widely used in feature extraction and clutter classification. The quality of clutter classification is often determined by how to accurately extract the CFT of each RDU from the original FT and more accurately characterize the features of each RDU and reduce the impact of noise and other uncertain factors [35]. In order to perform feature extraction on the original FT, as shown in Fig. 4, the main steps of the TDAO algorithm proposed in this article are listed as follows: 1) based on the similarly weighted between the same type of FTs by the training data and the nonhomogeneous FT penalty weighted, construct the weighting function G of arbitrary training FTs $\mathcal{X}_i$ and $\mathcal{X}_j$; 2) optimize each projection matrix $\boldsymbol{U}_m \in \mathbb{R}^{J_m \times I_m}$, $m = 1, 2, \ldots, M$, in turn, using an alternating optimization algorithm, and obtain the projection matrices after multiple iteration calculations; and 3) project the FT to obtain the CFT estimation value of each RDU.

The weighting function $G(i,j)$ between arbitrary FTs $\mathcal{X}_i$ and $\mathcal{X}_j$ is donated as

$$G(i,j) = \begin{cases} e^{-(\|\mathcal{X}_i - \mathcal{X}_j\|_F^2/2\sigma^2)}, & L_i = L_j \\ e^{-(\|\mathcal{X}_i - \mathcal{X}_j\|_F^2/(2\sigma^2))} - 1, & L_i \neq L_j \end{cases} \tag{13}$$

where $\|\mathcal{X}_i - \mathcal{X}_j\|_F^2$ is the tensor Frobenius-norm (see the Appendix), and L_i and L_j are the clutter categories of $\mathcal{X}_i$ and $\mathcal{X}_j$, respectively. σ is a scale parameter to characterize the similarity of FTs (in this article $\sigma = 1$). It is obvious that, for the same kind of FTs, the higher the similarity between $\mathcal{X}_i$ and $\mathcal{X}_j$ is, the closer the weighting function $G(i,j)$ is close to 1. In contrast, for heterogeneous FTs, the lower the similarity between $\mathcal{X}_i$ and $\mathcal{X}_j$ is, the weighting function $G(i,j)$ is close to -1.

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We assume that the number of training samples is N. With the maximum margin criterion, the objective function can be constructed as follows:

$$\operatorname{argmin}_{\mathcal{Y}_1,\ldots,\mathcal{Y}_N}\left\{\sum_{i,j}^{N} G(i,j)\cdot\left\|\mathcal{Y}_i-\mathcal{Y}_j\right\|_F^2\right\}. \tag{14}$$

The objective function can be written by (12) as follows:

$$\operatorname{argmin}_{\boldsymbol{U}_1,\ldots,\boldsymbol{U}_m}\left\{\sum_{i,j}^{N} G(i,j)\cdot\left\|(\mathcal{X}_i-\mathcal{X}_j)\times_1\boldsymbol{U}_1\times_2\boldsymbol{U}_2\times_3\cdots\times_M\boldsymbol{U}_M\right\|_F^2\right\}. \tag{15}$$

The TDAO algorithm optimizes another linear projection matrix $\boldsymbol{U}_m$ by fixing another $M-1$ projection matrices and then obtains the final M projection matrices by alternating optimization. Therefore, the objective function to optimize the mth projection matrix can be written as follows:

$$\operatorname{argmin}_{\boldsymbol{U}_m}\left\{\sum_{i,j}^{N} G(i,j)\cdot\left\|\begin{array}{c}(\mathcal{X}_i-\mathcal{X}_j)\times_1\boldsymbol{U}_1\times\cdots\times_{m-1}\boldsymbol{U}_{m-1}\\ \times_{m+1}\boldsymbol{U}_{m+1}\times\cdots\times_M\boldsymbol{U}_M\end{array}\right\|_F^2\right\} \tag{16}$$

where $\mathcal{Z}_i=\mathcal{X}_i\times_1\boldsymbol{U}_1\times\cdots\times_{m-1}\boldsymbol{U}_{m-1}\times_{m+1}\boldsymbol{U}_{m+1}\times\cdots\times_M\boldsymbol{U}_M$ is donated for succinctly expressing, so the CFT to optimize the mth projection matrix is $\mathcal{Y}_i=\mathcal{Z}_i\times_m\boldsymbol{U}_m$. The n-mode unfolding of a tensor $\mathcal{X}$ is denoted by $\mathcal{X}\Rightarrow_n\boldsymbol{X}^{(n)}$ (see the Appendix). We can transform the tensor norm into the matrix norm by the tensor property $\|\mathcal{Z}_i\times_m\boldsymbol{U}_m\|_F=\|\boldsymbol{U}_m\boldsymbol{Z}_i^{(m)}\|_F$. Therefore, we can rewrite (16) as follows:

$$\begin{aligned}&\operatorname{argmin}_{\boldsymbol{U}_m}\left\{\sum_{i,j}^{N} G(i,j)\cdot\|(\mathcal{Z}_i-\mathcal{Z}_j)\times_m\boldsymbol{U}_m\|_F^2\right\}\\&=\operatorname{argmin}_{\boldsymbol{U}_m}\left\{\sum_{i,j}^{N} G(i,j)\cdot\left\|\boldsymbol{U}_m\left(\boldsymbol{Z}_i^{(m)}-\boldsymbol{Z}_j^{(m)}\right)\right\|_F^2\right\}.\end{aligned} \tag{17}$$

With the relationship $\|\boldsymbol{X}\|_F^2=Tr(\boldsymbol{X}\boldsymbol{X}^T)$ between the matrix norm and the matrix trace, (17) can be expressed as follows:

$$\begin{aligned}&\operatorname{argmin}_{\boldsymbol{U}_m}\left\{\sum_{i,j}^{N} G(i,j)\cdot Tr\left[\boldsymbol{U}_m\left(\boldsymbol{Z}_i^{(m)}-\boldsymbol{Z}_j^{(m)}\right)\left(\boldsymbol{Z}_i^{(m)}-\boldsymbol{Z}_j^{(m)}\right)^T\boldsymbol{U}_m^T\right]\right\}\\&=\operatorname{argmin}_{\boldsymbol{U}_m}\left\{Tr\left\{\boldsymbol{U}_m\sum_{i,j}^{N} G(i,j)\cdot\left[\left(\boldsymbol{Z}_i^{(m)}-\boldsymbol{Z}_j^{(m)}\right)\left(\boldsymbol{Z}_i^{(m)}-\boldsymbol{Z}_j^{(m)}\right)^T\right]\boldsymbol{U}_m^T\right\}\right\}\end{aligned} \tag{18}$$

where $\boldsymbol{H}_m=\sum_{i,j}^{N} G(i,j)\cdot[(\boldsymbol{Z}_i^{(m)}-\boldsymbol{Z}_j^{(m)})(\boldsymbol{Z}_i^{(m)}-\boldsymbol{Z}_j^{(m)})^T]$ is donated for succinctly expressing, $Tr(\cdot)$ is the matrix trace operation, and the objective function can be abbreviated as follows:

$$\arg\min_{\boldsymbol{U}_m}\left\{Tr\left\{\boldsymbol{U}_m\boldsymbol{H}_m\boldsymbol{U}_m^T\right\}\right\}. \tag{19}$$

The above optimization problems can be converted to an eigenvector problem [36]. The projection matrix $\boldsymbol{U}_m\in\mathbb{R}^{J_m\times I_m}$ consists of eigenvectors corresponding to the first J_m large eigenvalues of $\boldsymbol{H}_m$. Finally, the final projection matrices are obtained by iterating and alternating optimizations, in turn, as $\boldsymbol{U}_m\in\mathbb{R}^{J_m\times I_m}$, $m=1,2,\ldots,M$, where the initial value of the projection matrix is selected as the identity matrix $\boldsymbol{U}_m=\boldsymbol{I}_m$, $m=1,2,\ldots,M$, and the order is I_m. The initial values of the projection matrices only have a small impact on the convergence of the proposed algorithm, which has been shown in Discussion (2). The iteration ending condition is that the error of the projection matrices between two iterations satisfies $\sum_{m=1}^{M}\|\boldsymbol{U}_m^{t_n}-\boldsymbol{U}_m^{t_n-1}\|_F<\varepsilon$, $t_n\geq 2$ or $t_n=T$, where T is the maximum number of iterations and ε is the iteration ending threshold parameter.

In fact, the original FT in this article is three-mode; only three projection matrices $\boldsymbol{U}_1$, $\boldsymbol{U}_2$, and $\boldsymbol{U}_3$ need to be optimized. The CFT is given as follows:

$$\mathcal{Y}_i=\mathcal{X}_i\times_1\boldsymbol{U}_1\times_2\boldsymbol{U}_2\times_3\boldsymbol{U}_3,\quad i=1,2,\ldots,N_RN_D \tag{20}$$

where the projection matrix $\boldsymbol{U}_1\in\mathbb{R}^{J_1\times I_1}$ reduces the range dimension from I_1 to J_1, $\boldsymbol{U}_2\in\mathbb{R}^{J_2\times I_2}$ reduces the Doppler dimension from I_2 to J_2, and $\boldsymbol{U}_3\in\mathbb{R}^{J_3\times I_3}$ reduces the feature dimension from I_3 to J_3. Finally, every CFT is extracted by projecting the original FT for each RDU.

D. Classification Based on AIR Distance

The extracted CFT contains the main features of each RDU. However, the classification methods generally ignore the correlated properties of different features directly by the Frobenius-norm distance between CFTs, which cannot make use of the distinctiveness of the extracted features. In this article, the CFT is unfolded along the feature dimension (mode 3) to obtain the feature matrix $\boldsymbol{Y}^{(3)}\in\mathbb{R}^{J_3\times J_1J_2}$, and the FCM corresponding to each RDU is obtained. Then, the clutter classification is performed under the AIR distance measurement between the FCMs.

The FCM corresponding to the CFT of the ith RDU is defined by

$$\boldsymbol{C}_i=\frac{1}{J_1J_2}\sum_{\xi=1}^{J_1J_2}(\boldsymbol{v}_\xi-\boldsymbol{\mu})(\boldsymbol{v}_\xi-\boldsymbol{\mu})^T \tag{21}$$

where $\boldsymbol{v}_\xi$ is the ξth column vector of the feature matrix $\boldsymbol{Y}^{(3)}$, and $\boldsymbol{\mu}=(1/J_1J_2)\sum_{\xi=1}^{J_1J_2}\boldsymbol{v}_\xi$ is the mean value of all column vectors.

It is easy to know from (21) that the FCM is a Hermite positive definite matrix formed by Riemannian manifolds, which does not satisfy the geometric properties of the Euclidean space. Therefore, the Euclidean distance cannot be used as a measure between FCMs. Considering the geometric characteristics of Riemannian manifolds, the similarity between FCMs can be measured by geodesic distance. For two Hermite positive definite matrices on the Riemannian manifolds, the geodesic distance commonly is the AIR distance [37] as

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follows:

$$D(\boldsymbol{C}_{\text{ref}}, \boldsymbol{C}) = \sqrt{\sum_{l=1}^{L} \ln^2 \lambda_l(\boldsymbol{C}_{\text{ref}}, \boldsymbol{C})} \tag{22}$$

where $\boldsymbol{C}_{\text{ref}}$ is the reference FCM of different regions (mainland clutter, azimuth ambiguous clutter, and sea clutter) and $\boldsymbol{C}$ is the FCM of arbitrary RDU. $\{\lambda_l(\boldsymbol{C}_{\text{ref}}, \boldsymbol{C})\}_{l=1,2,\ldots,L}$ is the generalized eigenvalues between $\boldsymbol{C}_{\text{ref}}$ and $\boldsymbol{C}$, L is the dimension of matrix $\boldsymbol{C}_{\text{ref}}$ or $\boldsymbol{C}$, and $\boldsymbol{x}_l \neq 0$ is the generalized eigenvector, which can be calculated as

$$\lambda_l \boldsymbol{C}_{\text{ref}} \boldsymbol{x}_l - \boldsymbol{C} \boldsymbol{x}_l = 0, \quad l = 1, 2, \ldots, L. \tag{23}$$

The TDAO-AIR clutter classification method proposed in this article can accurately extract and employ the correlation coefficient information, interferometric phase information, and spatial structural information in multiple directions for multichannel SAR images. It is more robust than the method only uses amplitude or phase information and can get better classification result by comparing the AIR distance between the FCMs.

E. Clutter Suppression Processing

To improve clutter suppression performance, we first need to obtain an accurate estimation $\hat{\boldsymbol{R}}_{cn}$ about CCM of the azimuth ambiguity region from the inshore ambiguity identification in our method. As a matter of fact, with the help of ambiguity identification, the training samples selected are close to IID. However, the contaminated clutter samples usually degrade the covariance estimate and reduce the detection performance when the targets exist. Thus, a preprocessing, such as the generalized inner product (GIP) method, is needed to prune the original training sample set contaminated by target-like signals in general [38]. Then, with the linear constraint minimum variance (LCMV) criterion [39], the optimal adaptive clutter suppression filter vector can be acquired by calculating the objective function as follows:

$$\begin{cases} \min_{\boldsymbol{w}} \ \boldsymbol{w}^H(v_r) \hat{\boldsymbol{R}}_{cn} \boldsymbol{w}(v_r) \\ \text{s.t. } \boldsymbol{w}^H(v_r) \boldsymbol{a}_s(v_r) = 1 \end{cases} \tag{24}$$

where $\boldsymbol{a}_s(v_r) = [1, \exp[j(4\pi v_r d_2/\lambda V)], \ldots, \exp[j(4\pi v_r d_K/\lambda V)]]^T$ is the multichannel airspace steering vector of the target with radial velocity v_r. The optimal solution of the objective function is $\boldsymbol{w}_{\text{opt}}(v_r) = (\hat{\boldsymbol{R}}_{cn}^{-1} \boldsymbol{a}_s(v_r) / \boldsymbol{a}_s^H(v_r) \hat{\boldsymbol{R}}_{cn} \boldsymbol{a}_s(v_r))$, where $(\cdot)^{-1}$ is the matrix inversion operation. This optimal weight is utilized to perform clutter suppression on azimuth ambiguous region, and a better clutter cancellation result is obtained.

The detailed process of the method proposed is divided into the following main six steps.

Step 1 *(Features' Description):* The correlation coefficients between channels of a multichannel SAR image are calculated, and the multilook interferometric complex image is obtained by multilook interferometric.

Step 2 *(Tensor Representation):* The correlation coefficient, interferometric phase, and spatial structure gradient features between channels are extracted to construct the FT for each RDU.

Step 3 *(Feature Extraction):* Select the FTs of training samples, construct the objective function, and use the proposed TDAO algorithm to get the projection matrices. Then, CFT is extracted by matrix projection of FT corroding to each RDU.

Step 4 *(Clutter Classification):* The CFTs are modularly expanded along the feature dimension to construct the FCMs, and the reference FCMs for different clutter regions is selected. Then, the AIR distance is used to measure the distance between each RDU's FCM and the reference FCMs, and further clutter classification is completed in the inshore region.

Step 5 *(Markov Random Field (MRF) Smoothing for Classification Result [40], [41]):* Since there may be isolated error points in the classification results, MRF is used to smooth the classification results.

Step 6 *(Clutter Suppression):* With the guide of the clutter classification result, the original multichannel clutter samples that subordinate to IID in the azimuth ambiguous region are selected for CCM estimation, the adaptive clutter weight vector is calculated, and the clutter cancellation result is obtained.

F. Computational Complexity Analysis

In this section, the computational complexity of the proposed method will be analyzed. Herein, an operation is first defined as one complex multiplication plus addition to describe the computational complexity. Assume that there are K azimuth receiving channels, and an SAR image contains the total RDUs of $N_R \times N_D$. The size of the multilook window is $W \times W$, each RDU is represented by an FT $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, the corresponding CFT is $\mathcal{Y}_i \in \mathbb{R}^{J_1 \times J_2 \times J_3}$, and the number of input samples is N. For convenience, we assume that $I = I_1 = I_2 = I_3$ and $J = J_1 = J_2 = J_3$. The number of iterations is T, and the class number is considered as C_{num} when an algorithm requires iterative processing. Considering the proposed processing framework, the computational complexity analysis can be divided into two parts: ambiguity identification and clutter suppression.

For the traditional methods, the single-channel methods generally have a computational efficiency. For example, the IMSR method identifies the azimuth ambiguity by calculating and judging the azimuth shifts of ambiguity with azimuth antenna pattern, which do not require complex multiplication plus additions. The AM&SF method determines the azimuth ambiguity by constructing a selective filter using Wiener filters, where the two Fourier transform operations are involved, and it requires $O[N_D \log(N_D)]N_R$ operations. For the MFAutoB method, it requires $W^2 K(N_R N_D)$ operations to estimate the multilook clutter covariance matrix. Then, the EM step consists of a matrix inversion operation (which requires $O[K^3]$ operations), a matrix determinant operation needing $O[K^3]$ operations, and a matrix product operation (which needs K^3 operations). With the T iterations, the cost of ambiguity identification of the MFAutoB method is $(2O[K^3] + K^3)C_{\text{num}} N_R N_D T + W^2 K(N_R N_D)$.

For the proposed method, in the ambiguity identification step, a multilinear product of FT requires $(2NJ)I^3$

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operations, a matrix product operation to construct the matrix $\boldsymbol{H}$ requires J^2I^2 operations, and a matrix eigenvalue decomposition (EVD), which corresponds to $O[4(1/3)(I)^3]$ complexity [42]. By alternately optimizing three projection matrices $\{\boldsymbol{U}_1, \boldsymbol{U}_2, \boldsymbol{U}_3\}$ and performing the T iterations operation, the cost of the feature extraction step is $\{O[4(1/3)(I)^3] + (2NJ)I^2 + J^2I^2\}(3T)$. Finally, the classification step based on AIR distance with different clutter classes requires $O[4(1/3)(J)^3]C_{\text{num}}$. Therefore, the operations of the proposed classification method are $\{O[4(1/3)(I)^3] + (2NJ)I^2 + J^2I^2\}(3T) + O[4(1/3)(J)^3]C_{\text{num}}$.

As for the clutter suppression step (the MFAutoB method and the proposed method are the same), we ignore the operations for removing the target-like signals. It first requires K^2L_s operations to estimate the CCM, where L_s is the number of the samples. Then, a matrix inversion operation (which requires $O[K^3]$), an optimal weight calculation needing $K^2 + K$ operations, and clutter cancellation (which requires $K(N_RN_D)$ operations) are performed in sequence. In summary, the computational complexity of the proposed method is

$$\left\{O\left[4\frac{1}{3}(I)^3\right] + (2NJ)I^2 + J^2I^2\right\}(3T) + O\left[4\frac{1}{3}(J)^3\right]C_{\text{num}} + K^2L_s + O[K^3] + K^2 + K + K(N_RN_D).$$

IV. Experimental Result and Analysis

In this section, the experiments on the simulated data and real data measured by the TerraSAR-X satellite are performed to evaluate the proposed method. To comprehensively demonstrate the advantages of the proposed method for azimuth ambiguity identification and suppression, some methods are performed for comparisons, including the AM&SF algorithm discussed in [10], the IMSR approach introduced in [11], the Tensor-FR method in [27], and the MFAutoB method proposed in [16] and [17].

A. Simulation Experiments

In this section, we utilize simulated space-borne SAR data to demonstrate the effectiveness of the proposed method. The simulation conditions are given as follows: the platform velocity $V = 7500$ m/s, the radar wavelength $\lambda = 0.0311$ m, the PRF PRF $= 1200$ Hz, and the antenna azimuth size is 10 m, which transmits radar signal with full aperture and is divided two receiving channels to receive at the same time. The EPCS between two received channels is $d_2 = 2.5$ m. The main-lobe land clutter noise ratio (CNR) is approximate to 23 decibels (dB), and the azimuth ambiguity CNR is approximate to 6 dB.

The simulated clutter scene is an island surrounded by the calm sea. Fig. 5 shows the imaging results of the simulated space-borne SAR data, which is obtained by multilook processing with a spatial window of 3×3. In Fig. 5(a), the white region in the middle represents the island, the left or right azimuth ambiguity is on both sides, and the sea clutter is on the marginal region. The multilook interferometric phase of the left or right azimuth ambiguous clutter [the red region and the blue region in Fig. 5(b)] is approximate to ± 2.5, while the theoretic mean of interferometric phase is

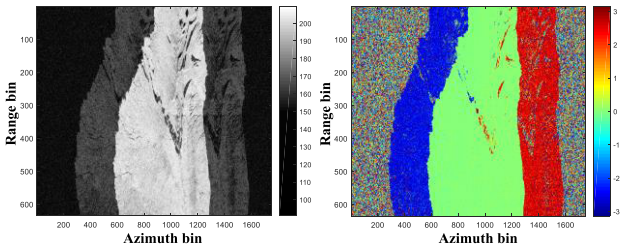


Fig. 5. Simulated space-borne SAR data. (a) Multilook interferometric amplitude. (b) Multilook interferometric phase. The colorbars represent the amplitude value or phase value.

$\theta_{\text{amb}} = \pm 2\pi \cdot \text{PRF} \cdot d_2/V \approx \pm 2.5$ in the azimuth ambiguous region. Therefore, it is obvious that our simulated space-borne SAR data are efficient and close to the real data. It is worth noting that the amplitude/phase characteristic of the shadow region of the land is the same as that of the ambiguity because it also is the ambiguous component of land clutter (the power of the local clutter is very small). Similarly, the shadow of the ambiguity is the two-order ambiguous component of land clutter, and it is close to noise for the antenna sidelobe weight. Therefore, it can be foreseen that our method will classify the shadow of the land region to the ambiguity and classify the shadow of the ambiguity region to the sea clutter.

From the results of the ambiguity determination of different methods in Fig. 6, the proposed method (we call the TDAO-AIR method) can well separate the mainland clutter, the azimuth ambiguous clutter, and the sea clutter, where the MRF spatial smoothing parameter is set to 2.5. At the same time, the boundaries between the categories are maintained as much as possible, as shown in Fig. 6(a). In order to show that our method can determine the azimuth ambiguity position and obtain the better clutter suppression performance, this article compares the Tensor-FR method (take the TDLA method for example in this article), the MFAutoB method (after the subsequent manual selection), the AM&SF algorithm, and the IMSR method, as shown in Fig 6(b)–(e). It is obvious that the Tensor-FR method has many misclassifications in the main-lobe clutter region and the azimuth ambiguity region. The AM&SF method wrongly divides a large number of sea clutter and mainland clutter into azimuth ambiguous clutter, and cannot accurately extract the azimuth ambiguous position from the inshore region; although the IMSR method can determine the location of the ambiguity clutter, it is necessary that the SAR image contains the main-lobe clutter corresponding to the azimuth ambiguity. The MFAutoB method could greatly improve the classification accuracy by the amplitude and phase features compared with the AM&SF method and the IMSR method, but it still cannot distinguish the shadow of the land region and the ambiguity region precisely. However, the TDAO-AIR method is an approach based on the actual echo data, which can accurately separate the different clutters and determine the azimuth ambiguity. In addition, our method utilizes the amplitude, phase, and spatial structure features comprehensively and can precisely classify the shadow region.

Fig. 7(b)–(f) shows the clutter suppression results on the azimuth ambiguous region of the TDAO-AIR method, the Tensor-FR method, the MFAutoB method, the AM&SF method, and the IMSR method, where the measurement in colorbars is in dB. Fig. 8 shows the clutter suppression

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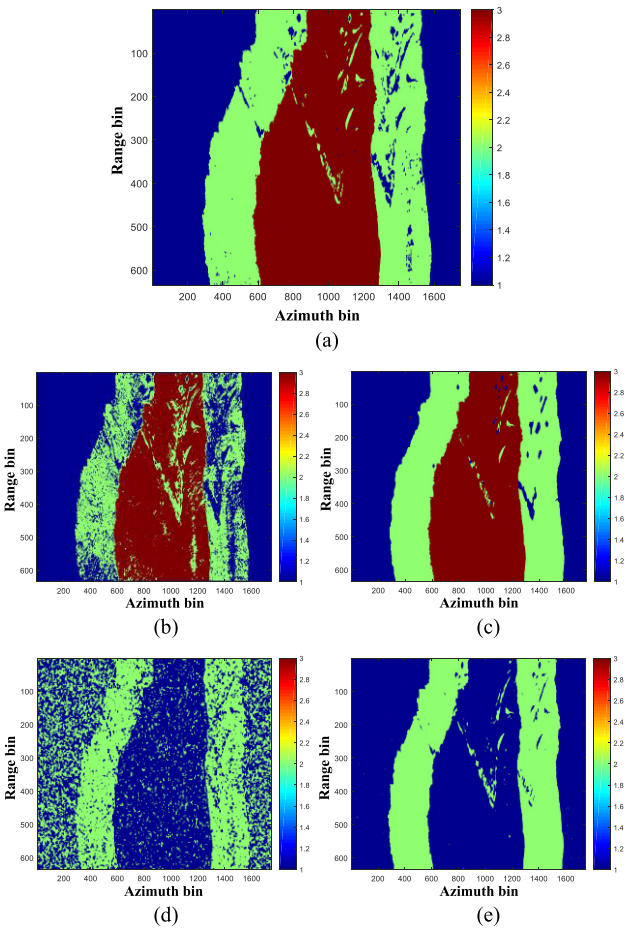


Fig. 6. Determinations of azimuth ambiguity by different methods. (a) TDAO-AIR method proposed. (b) Tensor-FR method. (c) MFAutoB method. (d) AM&SF method. (e) IMSR method.

cancellation result of different methods in range bin 300. The AM&SF method has a lot of sea clutter and mainland clutter mixed into the azimuth ambiguous clutter, and the clutter suppression performance has seriously deteriorated; The IMSR method cannot remove the part of the shaded region in the azimuth ambiguous clutter, and the amplitude/phase characteristics of the training samples still do not have the same distribution, which leads to a certain loss about the azimuth ambiguous clutter suppression performance; The Tensor-FR method has a certain deterioration in the clutter suppression performance because of some misclassifications for different clutter regions. The MFAutoB method and the TDAO-AIR method can select precisely azimuth ambiguous clutter samples that are subject to the same amplitude/phase statistical characteristics to estimate the CCM and construct the optimal weight, thereby obtaining the better clutter suppression performance for azimuth ambiguity. However, the MFAutoB method cannot effectively determine the number of classified clutter categories, which requires subsequent manual selection.

B. Experiments of TerraSAR-X Data

In this section, the real measured data near the Donghai Bridge in Shanghai are utilized to demonstrate the effectiveness of the proposed method again, which is collected by TerraSAR-X satellite on May 11, 2010. The TerraSAR-X satellite works in the StripMap mode and the HH polarization mode, which has an antenna with an azimuth size of 4.8 m

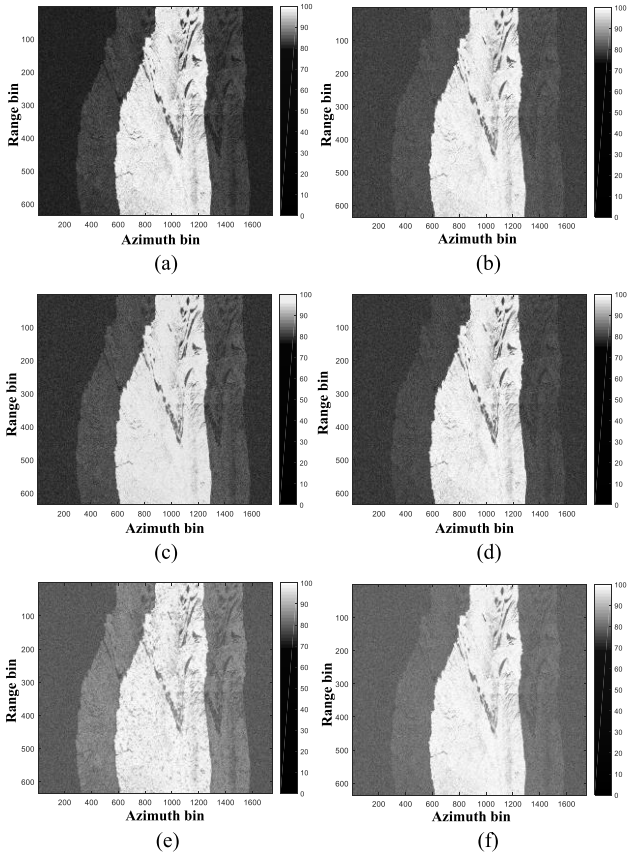


Fig. 7. Result of azimuth ambiguity suppression by different methods. (a) Original clutter power. (b) Proposed TDAO-AIR method. (c) Tensor-FR method. (d) MFAutoB method. (e) AM&SF method. (f) IMSR method.

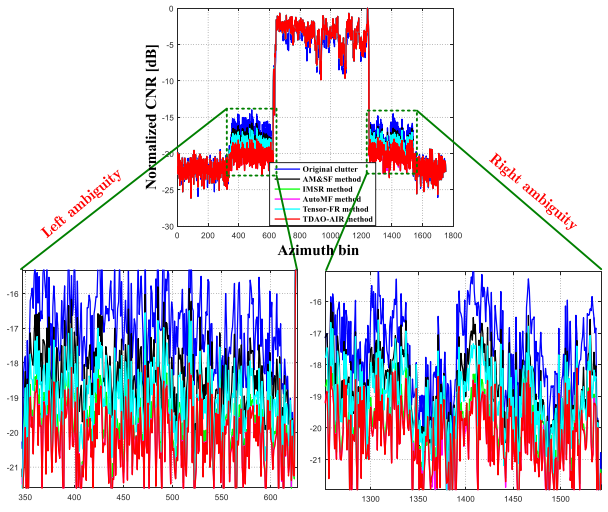


Fig. 8. Comparisons of performance for azimuth ambiguity suppression in 300 rang bin with different methods. The blue line represents the original clutter power. The black, green, pink, cyan, and red lines represent the suppression residual power of the AM&SF methods, the IMSR method, the MFAutoB method, the Tensor-FR method, and the proposed TDAO-AIR method.

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TABLE I
SYSTEM PARAMETERS FOR REAL

Parameters	Values
Wavelength	0.0311m
PRF	3.2274kHz
Channel Number	2
Baseline Length	2.4m
Velocity of Platform	7.6880km/s

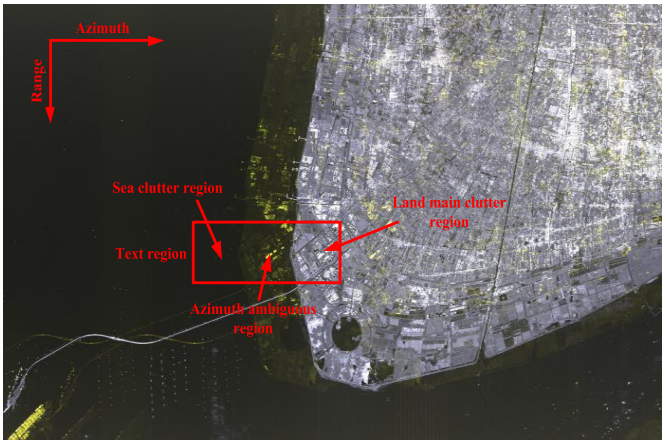


Fig. 9. Overview of TerraSAR-X data [German Aerospace Center (DLR)] and text region.

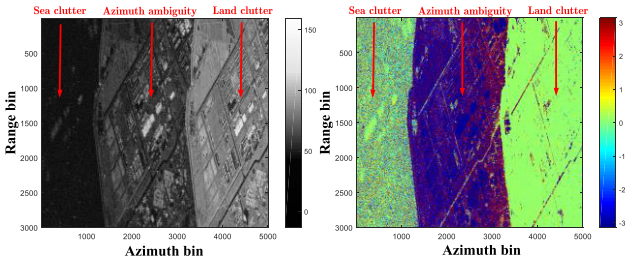


Fig. 10. SAR data of text region. (a) Multilook interferometric amplitude. (b) Multilook interferometric phase. The colorbars represent the amplitude value or phase value.

divided into two receiving channels. The spatial resolution is 1.177 m × 3.3 m (slant range × azimuth), and other radar system parameters are listed in Table I.

The test region selected in this article is shown in the red rectangle box, as shown in Fig. 9. According to the radar system parameters, the EPCS between channels is $d_2 = 1.2$ m. Therefore, the interferometric phase theoretical mean of the mainland clutter is $\varphi_{\text{lamd}} \approx 0$; the interferometric phase theoretical value of the left or right first-order azimuth ambiguous is $\varphi_{\text{amb}} = \pm 2\pi \cdot \text{PRF} \cdot d_2 / V \approx \pm\pi$. In this article, the multilook interferometric amplitude and phase images are obtained by a spatial window of 7 × 7, as shown in Fig. 10(a) and 10(b). The measurement in the colorbar of Fig. 10(a) is in dB, and it is a numerical value in Fig. 10(b). It is obvious that, on the test region, the interferometric phase of the mainland clutter is mainly concentrated in the vicinity of 0, and the interferometric phase of the azimuth ambiguous clutter is

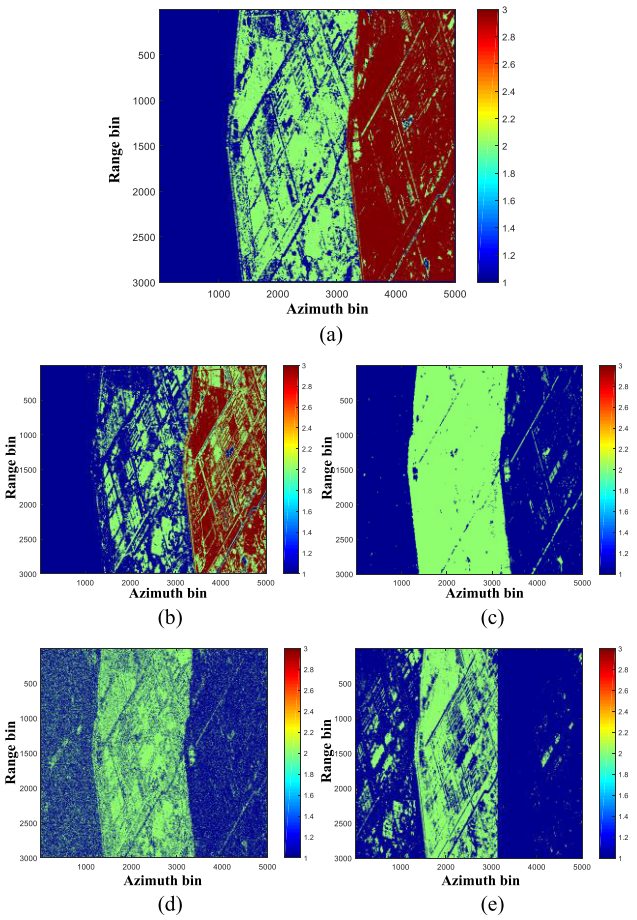


Fig. 11. Determinations of azimuth ambiguity by different methods. (a) TDAO-AIR method proposed. (b) Tensor-FR method. (c) MFAutoB method. (d) AM&SF method. (e) IMSR method.

concentrated on the vicinity of $-\pi$, which are consistent with the theoretical values. Like the simulation experiments, it is predictable that the proposed method can determine and suppress the azimuth ambiguous effectively.

Fig. 11 shows the results of different methods for clutter classification and location about azimuth ambiguous regions in the test region, where the MRF spatial smoothing parameter is set to 2.5. As shown in Fig. 11(a), our method divides azimuth ambiguous clutter, mainland clutter, and sea clutter very well and can accurately determine azimuth ambiguous location. Because the Tensor-FR method ignores the correlated properties of different features in the subsequent classification step, some main-lobe clutters are misclassified as the ambiguity clutter, and some azimuth ambiguity clutters are misclassified as the sea clutter. The MFAutoB method (after the subsequent manual selection) cannot classify the shadow of the land region and the ambiguity region precisely, as shown in Fig. 11(c). The classification results of single-channel methods, including the AM&SF and IMSR methods, for the text region are shown in Fig. 11(d) and (e), respectively. AM&SF not only misclassifies a large number of land and sea clutter but also has insufficient classification accuracy for azimuth ambiguous clutter, which makes it difficult to determine the azimuth ambiguity. As for the IMSR method, since some of the signal

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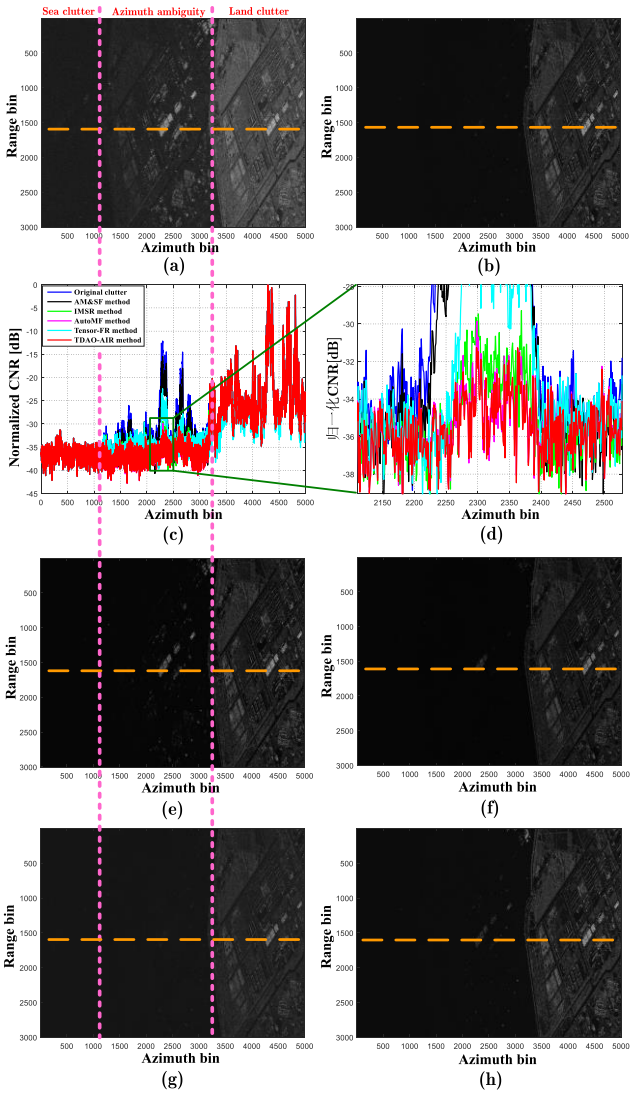


Fig. 12. Results and comparisons of performance for azimuth ambiguity suppression with different methods. (a) Original clutter. (b) Proposed TDAO-AIR method. (c) Normalized CNR for clutter before and after suppression in range bin 1600, where blue, black, green, pink, red, and cyan lines represent the corresponding lines in (a), (e), (f), (g), (b), and (h). (d) Results of the rectangle in (c). (e) AM&SF method. (f) IMSR method. (g) MFAutoB method. (f) Tensor-FR method.

sources (mainland clutter) that produce azimuth ambiguity are not in the test region, it cannot accurately identify the location of azimuth ambiguity. In other words, the IMSR method is invalid for this text region.

To further compare the ability of different methods to suppress azimuth ambiguity, this article first estimates the CCM of the ambiguity by the ambiguous clutter samples on the clutter classification map from different methods. Then, we perform clutter suppression on the azimuth ambiguity region and get the clutter suppression results, as shown in Fig. 12(b) and (e)–(h). Fig. 12(c) shows a normalized CNR comparison before and after clutter suppression in the range bin 1600, where the local magnification map on the azimuth ambiguity region is shown in Fig. 12(d). It is obvious that the AM&SF method, the IMSR method, and the Tensor-FR

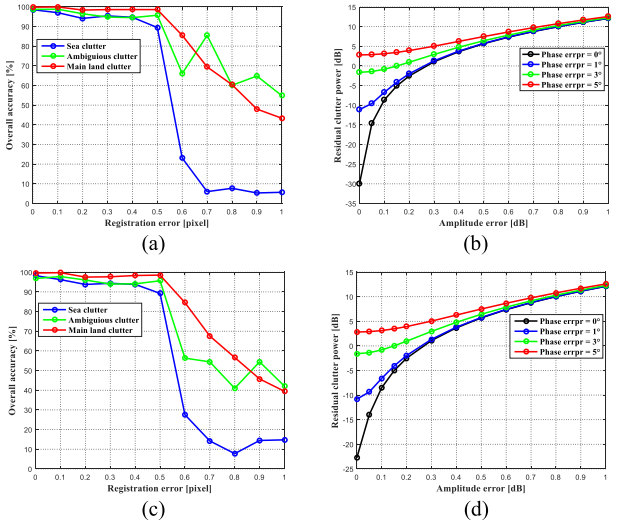


Fig. 13. Impact analysis of channel mismatch. (a) OA of simulation data with registration error. (b) Residual clutter power of the simulation data after suppression with amplitude/phase errors. (c) OA of the TerraSAR-X data with the registration error. (d) Residual clutter power of the TerraSAR-X data after suppression with amplitude/phase errors.

method have more clutter suppression residual, while the MFAutoB method and the TDAO-AIR method both have stronger clutter suppression ability and lower residual energy for azimuth ambiguity.

C. Discussion

1) Impact of Channel Mismatch: In this section, the impact of channel mismatch for the proposed method is discussed based on the framework shown in Fig. 1. First, it should be emphasized that the multiple receiving channels are assigned at the same platform in this article, so the channel imbalances are mainly caused by the effective phase center separation (EPCS) error, the beam pointing error, and the channel responding error. The EPCS error and the beam pointing error mainly affect the image registration, and the channel responding error mainly causes the amplitude/phase error of different receiving channels.

With the EPCS error and the beam pointing error, an inevitable shift, in both range and Doppler directions, will severely degrade the coherence of SAR images between channels and even make image registration failed. When the image registration error is larger than 0.5 pixels, the overall accuracy (OA) of different clutter regions will deteriorate sharply, as shown in Fig. 13(a) and (c). However, it should be emphasized that the multiple receiving channels are assigned at the same platform in this article, so the channel mismatch is easy to correct. In addition, in the current SAR-GMTI systems, such as the TerraSAR-X system [43] and the RADARSAT-2 system [44], the channel mismatch can be decreased as much as possible by the channel errors compensation or correction. By the precise orbit control and geometric registration operation, the SAR image registration accuracy can achieve 0.1 pixels in the StripMap mode of the TerraSAR system, which almost has no effect on the interferometric characteristic between images [45]. Meanwhile, the methods are

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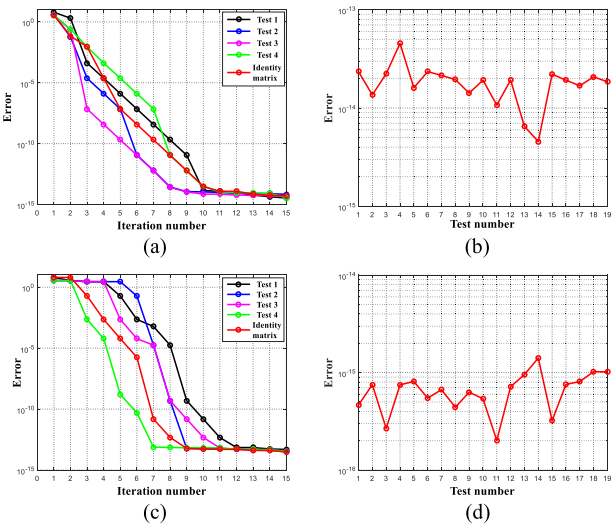


Fig. 14. Impact analysis of the initial value of projection matrixes. (a) Error between two iterations of simulation data. (b) Error between initial test matrices and density matrices after 12 iterations of simulation data. (c) Error between two iterations of the TerraSAR-X data. (d) Error between initial test matrices and density matrices after 12 iterations of the TerraSAR-X data.

based on the SAR echo data, including the azimuth cross correlation method [46], [47], the multiple signal classification techniques [48], the minimum entropy-based method [49], and so on, which can effectively reduce the impact of the channel mismatch. Thus, the correlation coefficient and the interferometric phase can be obtained. In addition, the clutter power of scenes also has an important effect on image registration. It is obvious that, if the clutter power is smaller, image registration is more difficult. Especially, for the ocean scene or the ocean containing moving ships, image registration will be a hard task. However, in this case, we do not need to classify the observation scene and even can perform the moving target detection without clutter suppression. As for the inshore area, the image registration can be performed with the strong land clutter.

As to the channel responding error, the internal calibration technology can be applied to reduce the system deviation by monitoring the instrumental drift within a system using calibration signals [50]. Bräutigam *et al.* [51] developed internal calibration for TerraSAR-X using pulse sequences, which recorded that the residual amplitude and phase errors are less than 0.16 dB and 2°. The residual amplitude/phase error usually affects the clutter suppression performance by deteriorating the correlation characters of different receiving channels. We test the residual clutter power after clutter suppression processing for different amplitude and phase errors when the original clutter power is preset. As shown in Fig. 13(b) and (d), the amplitude and phase errors are bigger, the residual clutter power will be more (the original clutter power is 30 dB, and the other system parameters are given in Section IV. However, when the gain (amplitude) error is less than 0.3 dB and the phase error is less than 3°, the residual clutter power is less than 3 dB with the system parameters in this article.

2) Impact of Initial Value of Projection Matrices: The impact of the initial value of projection matrices is discussed in the Appendix. For every test, we obtain the orthogonal matrices by extracting the eigenvectors of an arbitrary positive matrix. In Fig. 14, the error analysis result of projection matrixes is showed. In Fig. 14(a)–(d), we can see that different initial values of projection matrices may load to that a small difference in the convergence of the proposed algorithm. However, compared with the density matrix, the error of the final $\boldsymbol{U}_m \in \mathbb{R}^{J_m \times I_m}$, $m = 1, 2, \ldots, M$, obtained is very small in different tests. Thus, the initial value of projection matrices cannot seriously affect the classification effectiveness of the proposed method.

V. Conclusion

This article proposes a multifeature TDAO-AIR classification approach for inshore ambiguity suppression with multichannel SAR systems. In this article, the tensor is used to describe the correlation coefficient, the interferometric phase between channels, and spatial structure gradient characteristics in different directions for multichannel SAR images in the inshore regions. The proposed TDAO algorithm optimizes the projection matrices and extracts the CFT for each RDU. Then, the CFTs are modeled along the feature dimension, and the FCM for each RDU is constructed to classify the inshore clutter by the AIR distance measure. Finally, with the help of the clutter classification and region determination results, the subject IID clutter samples are selected to estimate CCM, and ambiguity suppression is further completed. The proposed method solves the problem that the strong ambiguous clutter in the near-shore water surface region causes the degradation of the clutter suppression performance of the multichannel SAR system. Compared with other methods, the main advantages of this method are the high classification accuracy and the great clutter suppression performance.

Appendix

The tensor algebra definitions that are relevant to this article are given as follows.

Definition 1 *(n-Mode Product):* The n-mode product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ and a matrix $\boldsymbol{U} \in \mathbb{R}^{J \times I_n}$ is denoted by $\mathcal{X} \times_n \boldsymbol{U}$. The elements of the n-mode product are defined by

$$(\mathcal{X} \times_n \boldsymbol{U})(i_1, i_2, \ldots, i_{m-1}, j_n, i_{m+1}, \ldots, i_M) = \sum_{i_n}^{I_n} \mathcal{X}(i_1, i_2, \ldots, i_M) \cdot \boldsymbol{U}(j_n, i_n). \quad (25)$$

Definition 2 *(Multilinear Product):* The multilinear product consists of the multiple application of n-mode products between tensor $\mathcal{X}$ and projection matrices $\{\boldsymbol{U}_m, m = 1, 2, \ldots, M\}$, which is represented as $\mathcal{X} \times_1 \boldsymbol{U}_1 \times_2 \boldsymbol{U}_2 \times_3 \cdots \times_M \boldsymbol{U}_M$.

Definition 3 *(n-Mode Unfolding):* The n-mode unfolding of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ is denoted by $\mathcal{X} \Rightarrow_n \boldsymbol{X}^{(n)}$, which is a transformation from tensor to the matrix. The unfolding matrix is denoted by $\boldsymbol{X}^{(n)} \in \mathbb{R}^{I_n (I_1 \times \cdots \times I_{n-1} \times I_{n+1} \times \cdots \times I_M)}$, where the column vectors of $\boldsymbol{X}^{(n)}$ are the n-mode vectors of tensor $\mathcal{X}$.

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Definition 4 *(Inner Product):* The inner product of two same size tensors $\mathcal{X}, \mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ is defined by

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{i_1=1}^{I_1} \sum_{i_2=2}^{I_2} \cdots \sum_{i_M=1}^{I_M} \mathcal{X}(i_1, i_2, \ldots, i_M) \cdot \mathcal{Y}(i_1, i_2, \ldots, i_M). \tag{26}$$

Definition 5 *(Frobenius Norm):* The Frobenius norm of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_M}$ is defined by $\|\mathcal{X}\|_F = (\langle \mathcal{X}, \mathcal{X} \rangle)^{1/2}$. The Frobenius-norm distance between tensor $\mathcal{X}$ and $\mathcal{Y}$ can be denoted by $D_F = \|\mathcal{X} - \mathcal{Y}\|_F$.

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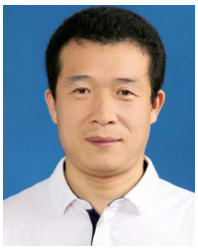
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